



General Chemistry I

Recitation 3

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Quiz

1. Ignoring tables of electronegativity values and guided only by the periodic table, arrange these atoms in order of increasing electronegativity: S, Cl, Sb, Se, In. Briefly explain your reasoning.
2. For each of the following atoms or ions, state the total number of electrons, the number of valence electrons, and the number of core electrons. (a) Ra^{2+} (b) Br (c) Bi^{2-} (d) Ga^+

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18										
															Prnctogens	Chalcogens	Halogens											
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight		C Solid											Metals		Metalloids	Nonmetals			2 He Helium 4.0026							
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122	Hg Liquid												Alkali metals		Alkaline earth metals	Lanthanoids	Transition metals	Post-transition metals		5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180	
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305	H Gas												Actinoids								13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798										
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29										
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71		72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)									
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)									

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
7	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

Bonds and molecular structure

Electrostatic force (ionic bonds)

Electron sharing (covalent bonds)

Delocalized electrons (metallic bonds)

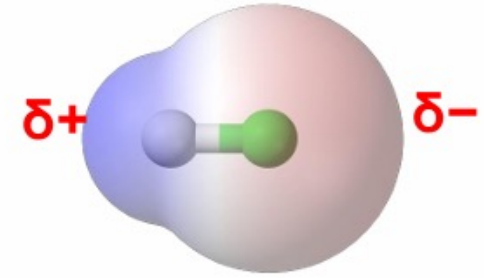
H-bonds?

π - π stacking?

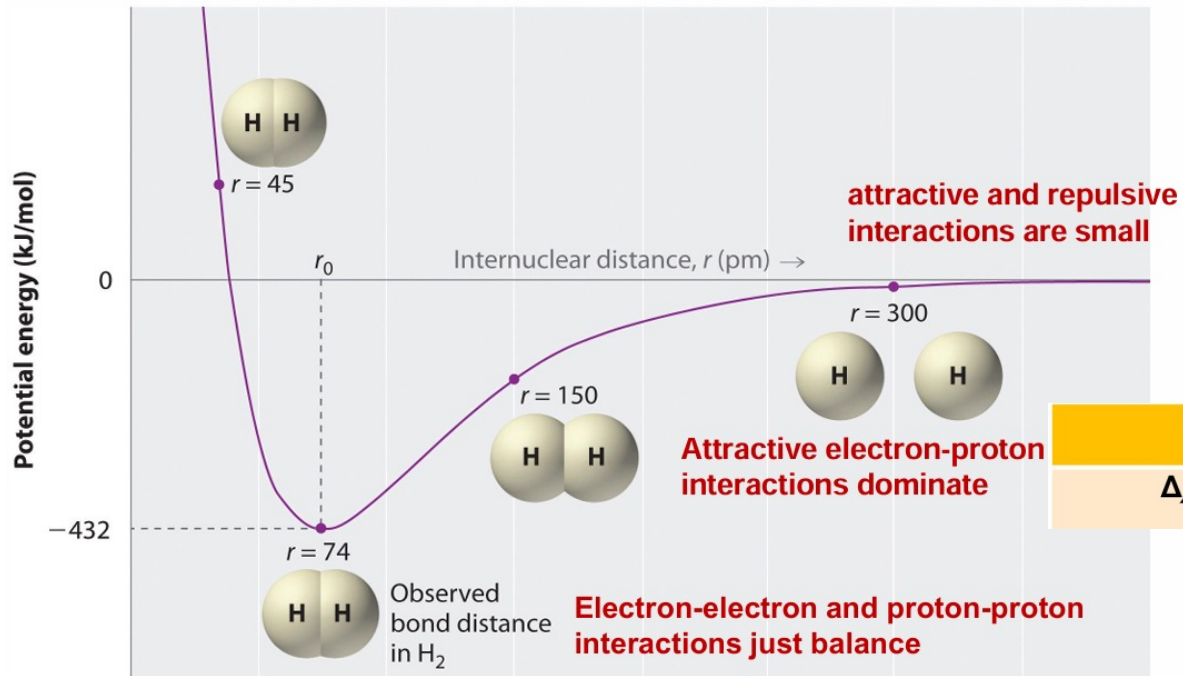
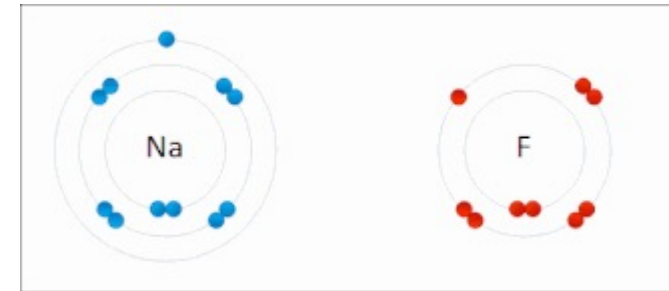
Van der Waals force?



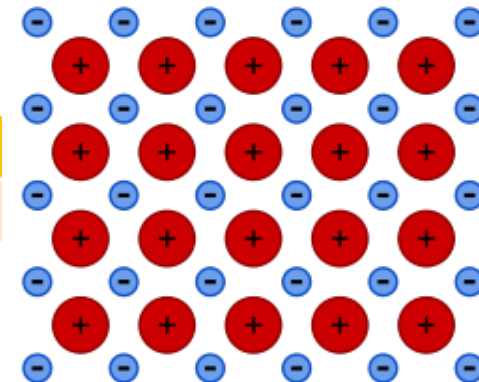
Not typical “chemical bonds”
But have similar properties



$$\Delta \propto q^2$$



Ionic	Polar Covalent	Covalent
$\Delta\chi > 2.0$		$\Delta\chi < 0.4$



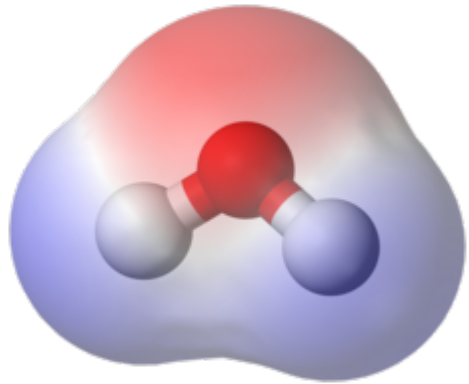
Bonds and molecular structure

Dipole moment (偶极矩) : **vector quantity**
electric dipole moment points **from the negative charge towards the positive charge**

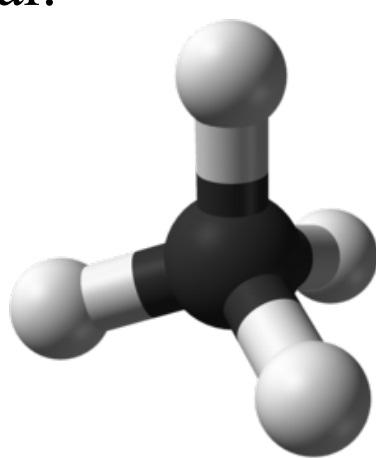
$$\mu = qR$$

If the bond dipole moments of the molecule do not cancel, the molecule is polar.

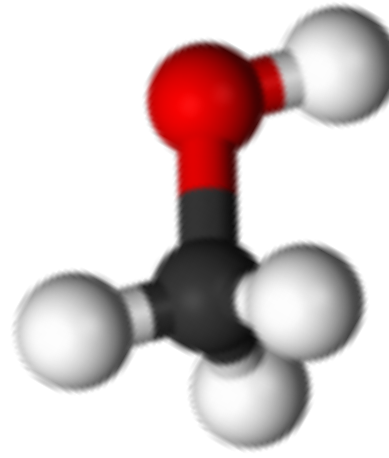
If the **vector sum** of the dipole moment is zero (cancel each other), the molecule is non-polar.



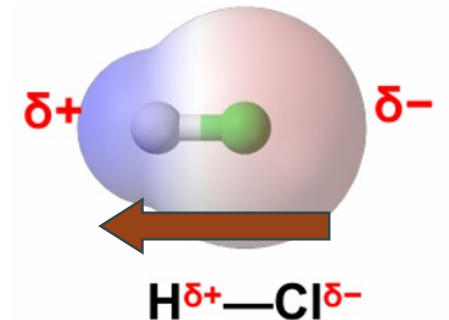
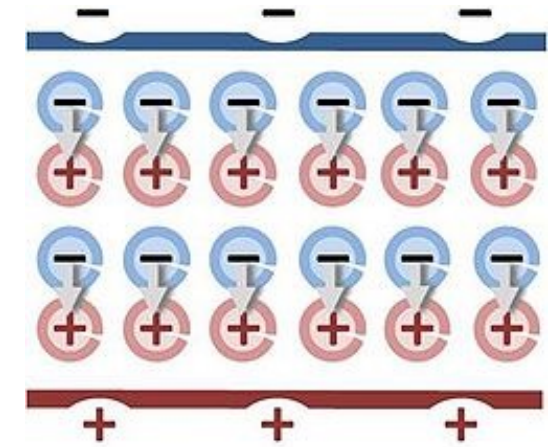
H_2O (polar)



CH_4 (non-polar)

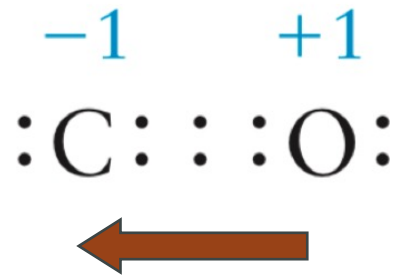


CH_3O (polar)



Bonds and molecular structure

Formal charge (形式电荷): the charge an atom in a molecule would have if the electrons in its Lewis diagram were divided equally among the atoms that share them.



Formal charge = number of valence electrons



- number of electrons in lone pairs

- $\frac{1}{2}$ (number of electrons in bonding pairs)

The less sum of FC(in absolute value), the more stable molecule is

Bonds and molecular structure

Lewis diagrams

Step 1: calculate total number of valence electrons + negative ion - positive ion

Step 2: calculate number of electrons needed to form noble-gas shell

Step 3: calculate numbers of shared electrons = step 2 - step 1

Step 4: assign two bonding electrons to each bond

Step 5: assign remaining bonding electrons to make double or triple bonds

Step 6: assign remaining electrons as lone pairs to the atoms, giving octets to all atoms except H.

Step 7: Determine the formal charge on each atom and write it next to that atom.

Step 8: If more than one diagram is possible, choose the one with the smallest magnitudes of formal charges and with any negative formal charges placed on the most electronegative atoms

Satisfying the octet rule

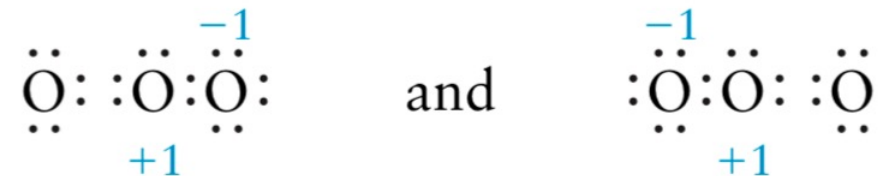
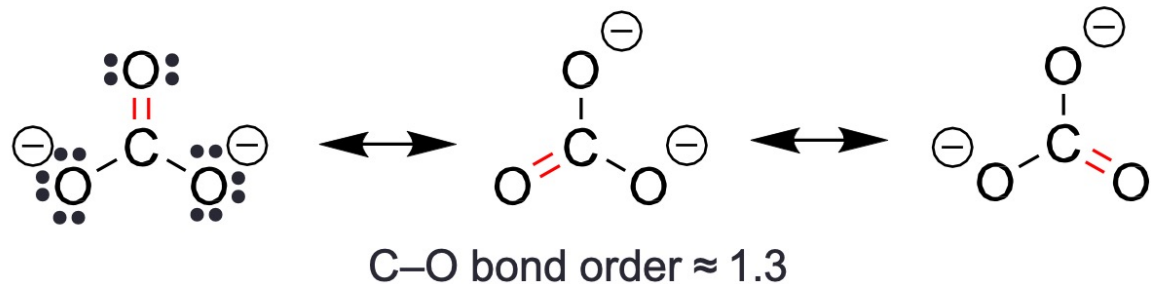
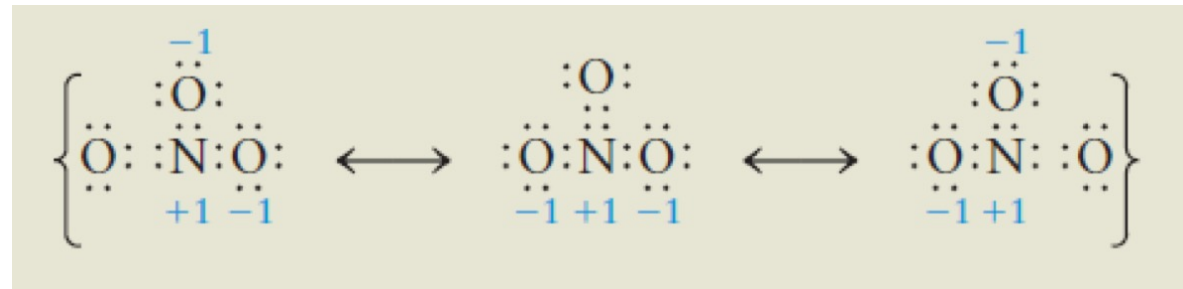
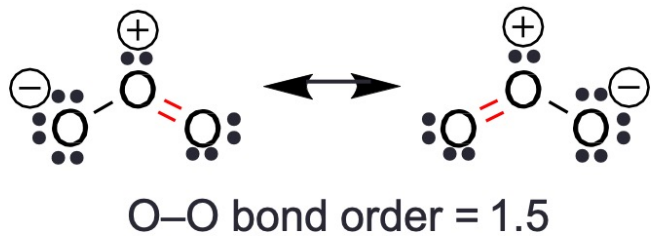
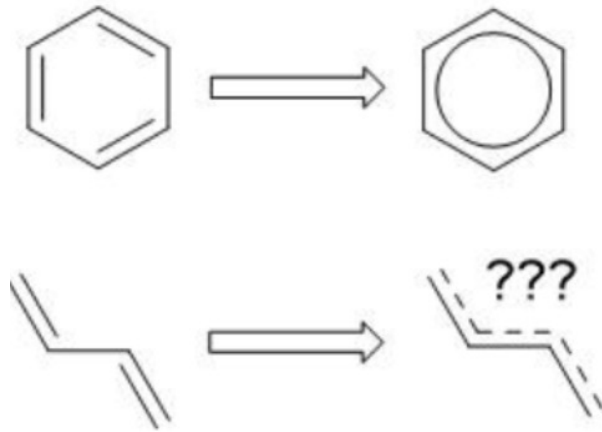
Octet rule works well for 2nd period elements.

Bonds and molecular structure

Resonance forms and bond order

Equivalent Lewis diagrams(in contribution) will be written as resonance forms

$$\text{B.O. (Resonance hybrid)} \approx \overline{\text{B.O. (Canonical forms)}}$$



Bonds and molecular structure

Lewis diagrams

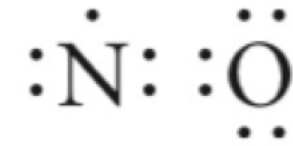
Hybrid orbital theory sounds better sometimes

$56(7 \times 8) - 48(6 + 6 \times 7) = 8$ bonding electrons

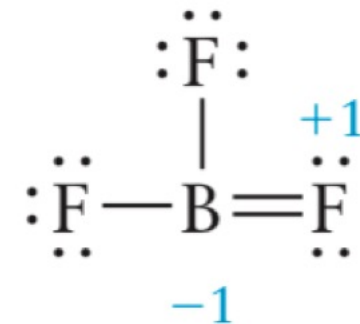
SF₆?

Step 6': abandon the octet rule for the central atom but preserve it for the terminal atoms. If any electrons still remain, assign them to the central atoms as lone pairs if the central atoms is not one from the 2nd period

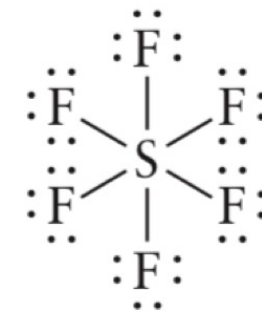
Molecular Orbital	Lewis Dot Structure
Bonding orbital	Shared electron pair
Nonbonding orbital	Lone electron pair
Delocalized orbital	Resonance structures
Two electrons from one atom	Coordinate bond
	Formal charge



Odd-electron?

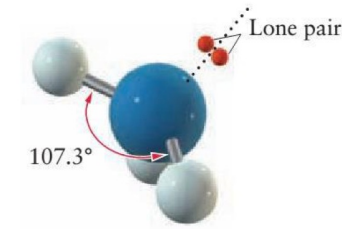
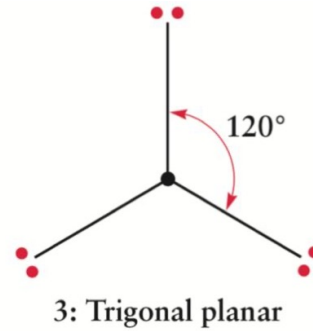
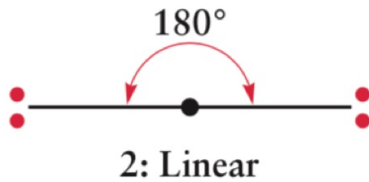


Delocalization π – *bonds*?

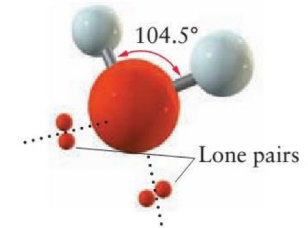


More than 4 terminal atoms?

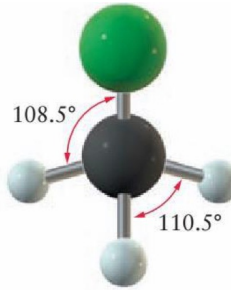
The molecular shape: Bond Order, Bond Angle, Bond Length



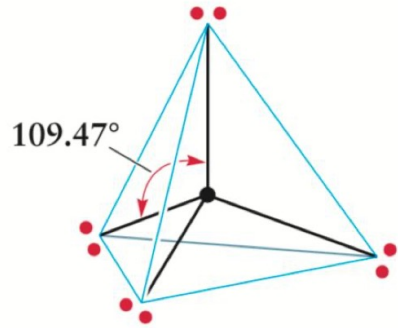
NH₃



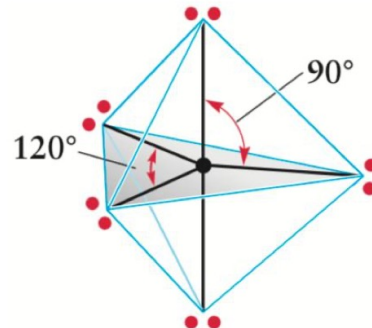
H₂O



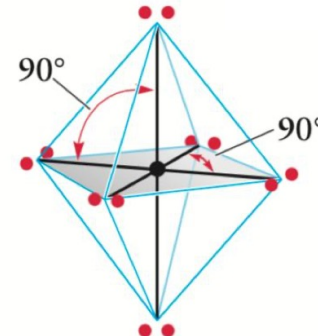
CH₃Cl



4: Tetrahedral



5: Trigonal bipyramidal

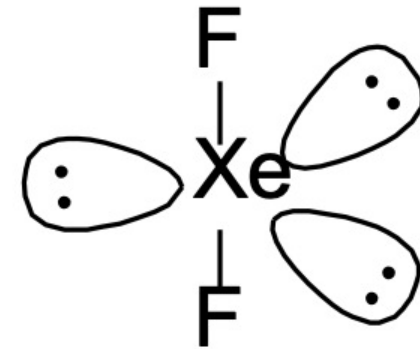


6: Octahedral

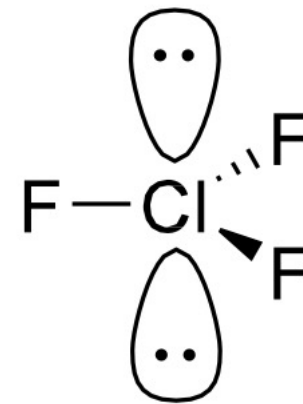
N-N
Single bond
Double bond
Triple bond

Bonds and molecular structure

SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX_2	Linear (线形)
3	AX_3	Trigonal planar (三角形)
	AX_2E	Bent (弯曲形)
4	AX_4	Tetrahedral (四面体)
	AX_3E	Trigonal pyramidal (三角锥)
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid (三角双锥)
	AX_4E	Distorted see-saw (跷跷板)
	AX_3E_2	Distorted T (T字)
	AX_2E_3	Linear
6	AX_6	Octahedral (八面体)
	AX_5E	Square pyramidal (四角锥)
	AX_4E_2	Square planar (正方平面)



Why linear?



Why planar triangle?

Bonds and molecular structure

Predict whether the molecules NH_3 and SF_6 will have dipole moments.

Solution

The NH_3 molecule has a dipole moment because the three N—H bond dipoles add to give a net dipole pointing upward to the N atom from the base of the pyramid that defines the NH_3 structure. The SF_6 molecule has no dipole moment because each S—F bond dipole is balanced by one of equal magnitude pointing in the opposite direction on the other side of the molecule.

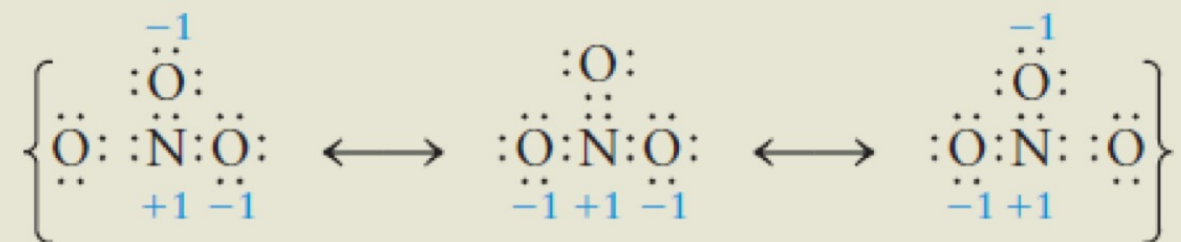
Bonds and molecular structure



Draw three resonance forms for the nitrate ion NO_3^- , and estimate the bond lengths.

Solution

NO_3^- contains 24 valence electrons. For each atom to have its own octet, $4 \times 8 = 32$ electrons would be required. Therefore, $32 - 24 = 8$ electrons must be shared between atoms, implying a total of four bonding pairs. These can be distributed in one double and two single bonds, leading to the equivalent resonance diagrams:



The two singly bonded oxygen atoms carry formal charges of -1 , and the charge of the nitrogen atom is $+1$. The bond lengths should all be equal and should lie between the values given in Table 3.5 for $\text{N}-\text{O}$ ($1.43 \text{ \AA}$) and $\text{N}=\text{O}$ ($1.18 \text{ \AA}$). The experimentally measured value is $1.24 \text{ \AA}$.